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Periodic Vortical Gust Encounter and Mitigation Using Closed Loop Control

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A wing's response to periodic vortical gusts of different frequencies and magnitudes was investigated and controlled using a PID based closed loop control system. All experiments were conducted in the University of Dayton Water Tunnel (UD-WaT). The vortical gust was generated using an oscillating flat plate pitched about its quarter chord. The frequency of the vortical gusts was controlled by the oscillation frequency of the flat plate. Simultaneous Time Resolved Particle Image Velocimetry (TR-PIV), and force measurements were collected to capture the vortical gust and its interaction with a downstream flat plate at a fixed angle of attack. The force histories were correlated with the flow physics to determine the cause of force transients. A PID-based closed loop control system was then implemented to mitigate the periodic vortical gusts. The flow field responsible for successful gust mitigation was studied using TR-PIV. The closed loop controller was able to achieve up to 63.32% average mitigation. PIV data revealed the effective angle of attack history for gust alleviation, and the error in $\mathbf{C_L}$ was higher at higher $\dot{\alpha}_{\mathrm{eff}}$.

Nomenclature

A	=	Gust generator oscillation amplitude
α	=	Angle of attack
$\dot{\alpha}$	=	Time derivative of angle of attack
c	=	Sensing flat plate chord
C_L	=	3D lift coefficient
C_{L_α}	=	C_L vs α lift curve slope
C_l	=	2D lift coefficient
k	=	Reduced frequency
ω	=	Flat plate circular frequency
u	=	x component of velocity
U_∞	=	Free stream velocity
v	=	y component of velocity
t*	=	Convective time

Abbreviations

AoA	=	Angle of Attack
DNS	=	Direct Numerical Simulations
GR	=	Gust Ratio
LE	=	Leading Edge
PID	=	Proportional Integral Derivative
TE	=	Trailing Edge
TR-PIV	=	Time Resolved Particle Image Velocimetry
UAV	=	Unmanned Aerial Vehicle
UD-WaT	=	University of Dayton Water Tunnel

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I. Introduction

The study of gust encounters has become increasingly important for many applications of aerodynamics. Birds [1], small scale UAVs, and wind turbines [2] all operate in unsteady environments, and a deeper understanding of their interactions with aerodynamic disturbances is of great interest to increase performance of aerodynamic devices. Disturbances where flow velocity changes in time can be broadly categorized as gusts, which are often divided into three categories: transverse, streamwise and vortical [3]. This study focuses on vortical gusts.

Transverse gust interactions and mitigation have been studied extensively by several researchers. Sedky et. al. [4] were able to effectively mitigate C_L transients present in a transverse gust encounter up to 50% using open loop pitching maneuvers based on simulations of closed loop control with a modified Goman Khrabrov model. This model proved to be effective in capturing changes in lift due to effective angle of attack changes, but it performed poorly in the gust recovery phase of the gust encounter due to the formation of large trailing and leading edge vortices. Andreu-Angulo and Babinsky [5] also achieved substantial mitigation up to 95% of a transverse "top hat" gust encounter. They used a custom designed open loop pitching profile based on potential flow theory with some manual corrections to achieve an effective AoA of zero.

A wide body of excellent work exists that studies vortical gust interactions with a variety of airfoils and flat plates at different Reynolds numbers. Direct Numerical Simulations (DNS) have found that the circulation of an incoming vortical gust is roughly proportional to the change in C_l through time of a symmetrical airfoil [7]. Additional numerical studies have also determined that for a symmetrical airfoil, there is typically a sign change in the C_l history on a wing as it interacts with a vortical gust. For a counterclockwise gust, there is an initial positive spike in C_l as the gust approaches the airfoil, followed by a smaller negative peak [7][8]. Water tunnel experiments performed by Medina et. al. [9] show that the same transients exist in water tunnel testing and that the signs of the spikes swap for an opposite rotation gust.

Vortical gust encounters are categorized as either transient or periodic [10]. Transient gusts are comprised of a single vortex that travels downstream to interreact with a wing. These are the type investigated by the research discussed above. In addition to transient gusts, periodic gusts also present important encounters where alternating clockwise and counterclockwise rotating gusts travel downstream at a given frequency. Medina et. al. [9] investigated both transient and periodic gusts. They found that periodic gusts produced a unique C_L response rather than resembling repetition of transient gust interactions. For periodic gusts, there is a repeating and alternating pattern in the C_L history.

A comparison between transverse and transient vortical gust encounters by Biler et. al. [10] found that there were several important similarities between the two gust types despite having several fundamental differences in their structure and generation. Both gusts resulted in large lift transients, variations in the gust induced AoA, and circulation shed from the flat plates. For the same gust ratio, the transverse gust had a steeper and higher lift transient and more shed circulation. The vortical gust, however, had a larger secondary peak while the plate in the transverse gust reattached almost immediately after exiting the gust.

Many strategies for generating vortical gusts have been developed in previous studies. Some designs use pitching airfoil type gust generators that generate start-stop vortices which convect downstream to generate vortical gusts in both wind [11][12] and water [13]. Hufstedler and McKeon also evaluated a heaving flat plate offset from the test article in its resting position to prevent the sensing wing from "seeing" the wake of the gust generator in between gust encounters.

The method proposed by Rockwood and Medina [14] was used to generate periodic vortical gusts of a desired strength and frequency. Rockwood and Medina [14] showed that by rotating a cylinder with an attached TE plate at different rates, the Strouhal number could be modified to change the rate vortices shed from the gust generator. By changing the amplitude of rotation of a cylinder with a trailing edge flat plate, the amplitude and frequency of vortices could be further controlled to simulate vortical gusts in a water tunnel. Using this type of cylinder-plate gust generator was considered for this study, but a flat plate design was chosen instead due to the size of the cylinder wake compared to a flat plate.

The primary challenges in open loop unsteady lift prediction are modeling the changes in α and $\dot{\alpha}$ accurately for successful gust mitigation. Since all the standard models assume attached flow, theoretical predictions of lift in unsteady flows cannot fully capture the flow transients and resulting effects at moderate to high AoA excursions. The work presented in this paper explores an alternate method to obtain the α and $\dot{\alpha}$ in vortical gust encounters by using a closed loop controller. Using a closed loop controller is beneficial because it does not necessarily require knowledge of a gust's composition a priori and therefore could be more broadly applicable.

Herrmann et. al. [11] were able to achieve 58.6% mitigation of a periodic vortical gust disturbance in wind tunnel experiments using a trailing edge flap with a Proportional Integral (PI) feedback and model-based feedforward closed loop controller. This study also investigated quasirandom gust encounters with the same controller and measured improved mitigation of 63.9%. It was determined that the feedforward capabilities of the controller were essential to perform meaningful mitigation.

The method proposed here suggests that a closed loop controller designed by Mongin et. al. [15] could be used to achieve a desired C_L through a vortical gust disturbance while the flow physics are recorded using TR-PIV. Mongin et. al. showed that a pitching flat plate with both PID and Full State Feedback based controllers could accurately track C_L profiles through a range of amplitudes and unsteadiness as measured by reduced frequency defined in Eq. (1)[16]. These profiles are analogous to a virtual gust, as opposed to the physical gusts studied here. This paper will use a specially tuned version of the PID controller to hold a constant C_L through various physical vortical gust encounters while recording time synchronized PIV data to understand the flow physics present in successful gust mitigation. By understanding these flow physics, future control methodologies can be tailored to address characteristic flow features improving the performance of aerodynamic devices that operate in gusty environments.

$$k = \frac{\omega c}{2U_\infty} \tag{1}$$

II.Experimental Setup

A. University of Dayton Water Tunnel

All experiments were performed at the University of Dayton Water Tunnel, shown in Fig. 1. The facility is a horizontal free-surface water tunnel with a 6:1 contraction and a 38 cm wide and 46 cm high test section of 152 cm length. The free-surface speed range is 7.5 cm/s to 38 cm/s. For this study, the freestream velocity of the tunnel was set to 0.28 m/s. The tunnel is equipped with a one degree of freedom motion stage consisting of a H2W Technologies linear motor driven by a Xenus XTL-230-18 motor driver from Copley Controls. This motor is outfitted with a rack and pinon gear that converts its linear motion to a rotational motion to pitch a test article in the tunnel test section. The tunnel is also equipped with an Applied Motion Products HT23-598DC-ZAC stepper motor and rotational controller used to rotate the gust generator.

Fig. 1 University of Dayton water tunnel facility

B. Sensing Flat Plate

All experiments were performed on an AR 2 semi-span flat plate made from borosilicate glass with a 3in. (0.076m) chord and 6in. (0.152m) span attached to a submersible ATI Mini40 IP68 force transducer. The chord-based Reynolds number was 24,119 using a freestream velocity of 0.28 m/s. The use of splitter plate ensures an effective AR of 4. The flat plate rotates about and was mounted to the force transducer at its quarter chord. The static lift curve of the flat plate is shown in Fig. 2 along with the AR corrected theoretical lift slope from the Helmbold Equation shown in Eq (2)[17].

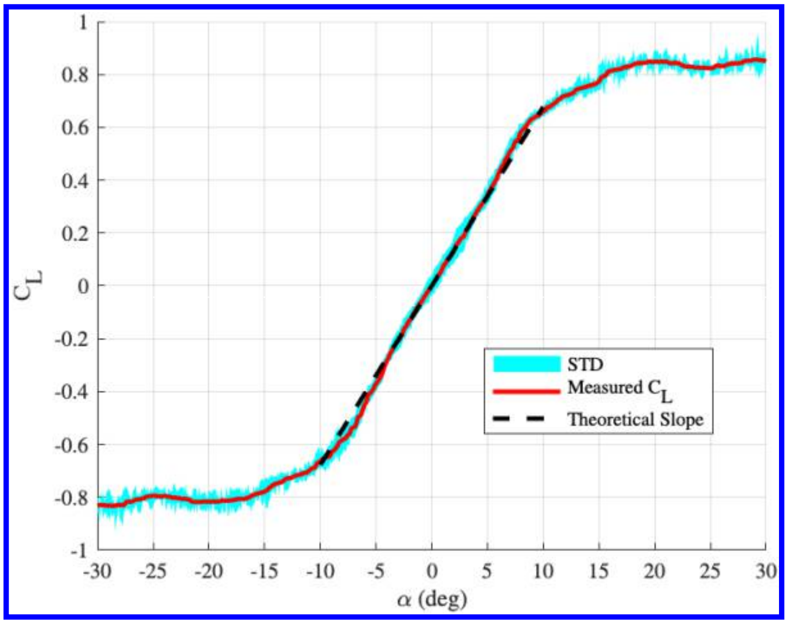

Fig. 2 Flat plate static lift curve

$$C_{L_\alpha} = \frac{\pi AR}{1 + \sqrt{1 + \left(\frac{AR}{2}\right)^2}} \tag{2}$$

C. Vortical Gust Generator

The vortical gust generator consists of a 3in. chord, 1/8in. thick, 18.5in. span aluminum flat plate that is pitched about its quarter chord. The gust generator extends from above the surface of the water down to nearly the bottom of the tunnel where it is supported by a shaft held to the bottom of the tunnel with a magnetic post. The quarter chord of the gust generator was placed 5 chords (15in.) upstream of the sensing flat plate measured from the pitch axes as shown in Fig. 3. This distance was chosen as a good middle ground to minimize the effect of the wake of the gust generator between gusts while keeping good vortex coherency as it convected downstream. The gust generator was pitched in a sinusoidal profile at either 15° or 30° amplitudes and a range of frequencies to generate several different periodic vortical gusts of varying strength and frequency.

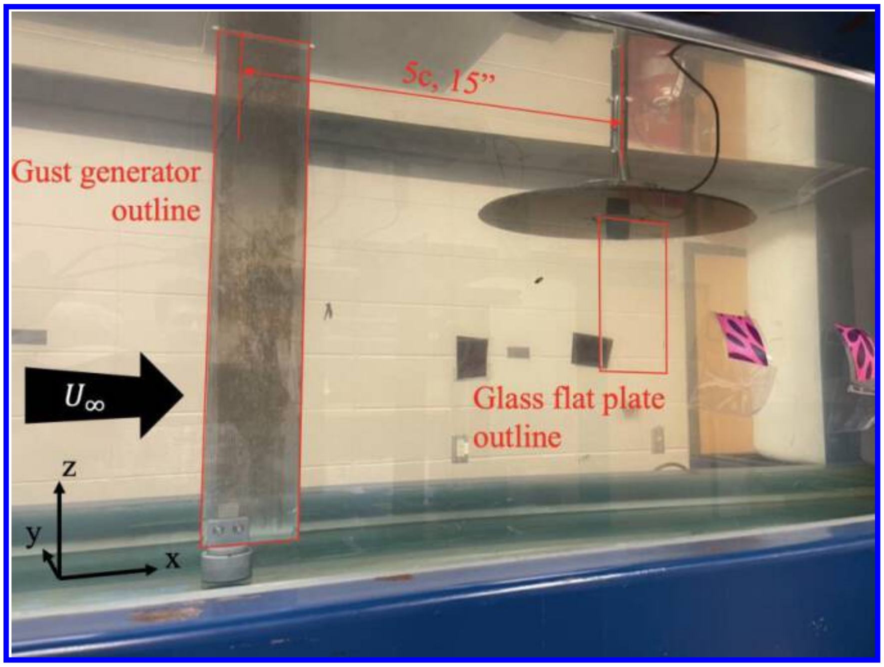

Fig. 3 Gust generator and flat plate in test section

D. dSPACE MicroLabBox

The closed-loop feedback and control for this system is performed with a dSPACE MicroLabBox using the MATLAB LabVIEW Real Time Interface (RTI) programming environment. The MicroLabBox was run at 1khz for these experiments. Force measurements were performed with an ATI Mini40 force transducer with NET F/T box interfaced with the MicroLabBox. The linear motor driver is connected to two analog outputs, one for enabling the motor and the other for commanding the current of the motor. Another connection between the linear motor driver and MicroLabBox is made via the digital I/O ports to receive information from the motor encoder. The stepper motor driver is controlled via another digital I/O port to send PWM step and direction signals to the stepper motor to command a desired position based on feedback from its shaft encoder. Both motors' positions are PID controlled with the MicroLabBox which were independently tuned to achieve a commanded position.

E. Control System

A simplified block diagram of the closed loop control system is shown in Fig. 4. A reference C_L is commanded and measured against the actual sensed C_L measured by the Mini40 force sensor in the circular difference block. The difference between these desired and measured signals is sent as the error to the PID controller. The PID gains of P=0.3, I=9, and D=0.001 were determined by manual tuning of the controller using both virtual and physical gusts. The controller outputs a commanded α which is achieved by the pitch motor. The plant in this case is the physical system of the flat plate and water in the tunnel which generate forces. The lift force is then measured by the force sensor and returned to the difference block closing the loop.

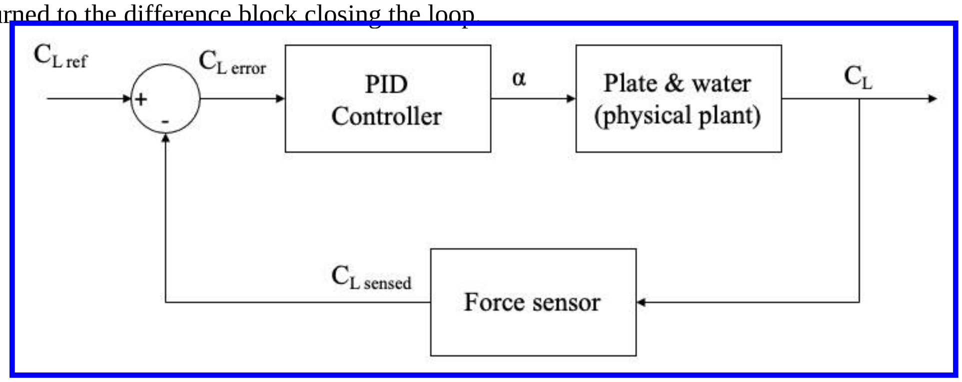

Fig. 4 Control system block diagram

F. Particle Image Velocimetry

PIV studies were performed using a PCO dimax S4 camera with a 2016x2016px CMOS sensor. 60μm polyamide particles were used to seed the tunnel and were illuminated with a 2.4W 532nm LaVision continuous laser. A sensitivity study was performed on the camera frame rate to ensure that particle displacement ranged from 8-10 pixels. The laser sheet was positioned perpendicular to the plate at the midspan. Due to the finite span of the plate, 3D effects are present, but they cannot be isolated with this setup. The glass construction of the flat plate allowed the laser to shine directly through it illuminating both the top and bottom surfaces. All PIV experiments were run for a total of 10 periods and phase averaged based on the frequency of the gust generator. Table 1 shows a test matrix of all the experiments completed for each of the gust characterization, fixed AoA open loop cases, and commanded C_L closed loop cases. For the open and closed loop cases, two settings were commanded for either zero C_L cases ($C_{L_{set=0}}$) or non-zero C_L cases ($C_{L_{set\neq 0}}$). In $C_{L_{set=0}}$ cases an open loop α=0° and closed loop C_L=0 were commanded. In $C_{L_{set\neq 0}}$ cases, α=6° and C_L=0.35 were commanded based on the static lift curve shown in Fig 2 to simulate realistic lifting conditions.

Table 1 PIV Test Matrix

Type	Amplitudes (deg)	k	AoA holds (deg)	CLs commanded
Gust Characterization	15, 30	0.3, 0.2, 0.1	N/A	N/A
Open Loop	15, 30	0.3, 0.2, 0.1	0, 6	N/A
Closed Loop	15, 30	0.3, 0.2, 0.1	N/A	0, 0.35

III. Results

A. Gust Characterization

Gust characterization experiments were conducted by oscillating the gust generator and collecting PIV data with no flat plate located downstream of the gust generator. This allowed the gust to be viewed without influence or obstruction by the flat plate. Fig. 5 shows the unwrapped gust canvases generated by the TR-PIV data. The temporal data has been rearranged in space to show the alternating gust profiles as they would appear upstream of the flat plate. The centerline where the flat plate is located is represented by the black long-short dashed line. Note how the x axis is different for each of the reduced frequencies due to varying period lengths. Gray vertical lines have been added at x/c=5 in each plot to give a sense of scale for how the period length shrinks with reduced frequency. The k=0.1, A=30° case yielded highly incoherent gusts and therefore was not considered valuable for this study. The gust characterization study revealed that the higher k=0.3 cases produced more visually coherent vortices while the lower k<0.3 cases produced less visually coherent vortices. These less coherent gusts still presented some interesting challenges for the controller to overcome which will be discussed in the sections below.

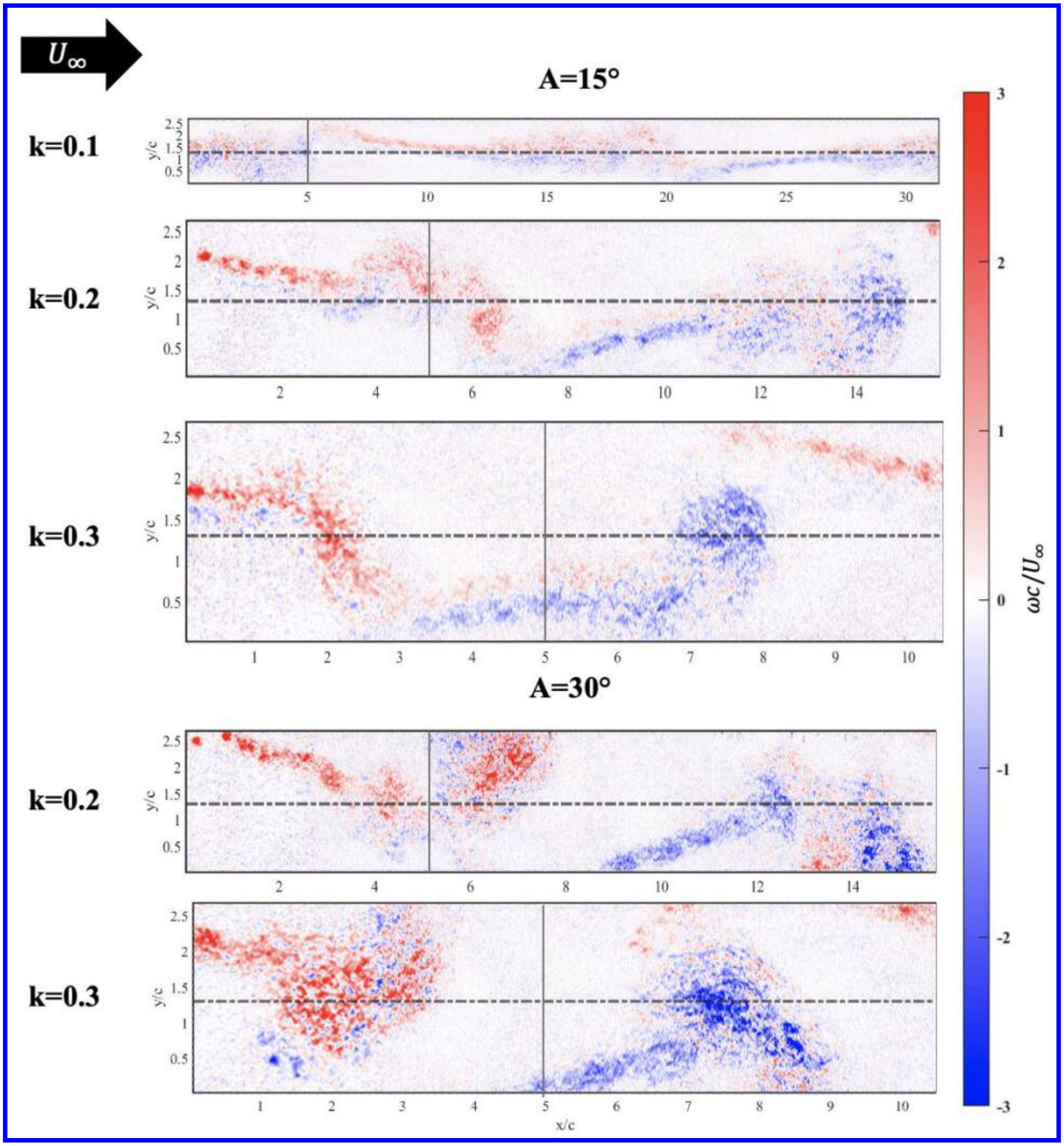

Fig. 5 Alternating vortical gust generated at various gust generator frequencies and amplitudes.

Fig. 5 shows how higher amplitude oscillation generates larger normalized vorticity values in the gusts as shown by the darker vorticity contours in the 30° cases than the 15°. For k=0.2, a higher amplitude also tended to result in larger departure of the vortex from the centerline. The gust generator and sensing plate were aligned in the freestream direction, but as the k=0.2 A=30° cases show, the vortices tended to move away from the centerline as they traveled downstream. The k=0.2 cases also contain a group of alternating vortices in close proximity. Especially in the A=30° case, the vorticity contours show how at about x/c=15, there is a leading blue negative vortex that is immediately followed by a red positive vortex at x/c=13, and finally a third negative vortex at about x/c=12. Further upstream, the opposite pattern is present with a red leading vortex at x/c=7 followed by a blue and another red vortex.

The induced AoA caused by the gust was measured by selecting a vertical line 0.33 chords in front of the flat plate as annotated in Fig. 6. These velocities were then used to obtain an instantaneous induced AoA using Eq. (3). The induced AoAs are plotted in Fig. 8 for the $C_{L_{Set}=0}$ cases and Fig. 9 for the $C_{L_{Set}\neq 0}$ cases. These induced angle of attack histories also show trends in gust coherency. The profiles show that the k=0.3 cases are comparatively less noisy than the 0.1 and 0.2 cases, indicating a greater coherency in the vortex. The 0.3 cases have clear peaks indicating where the core of the vortex passes, and the induced angle of attack flips sign across the vortex. Further, the slopes on either side of this peak are clearly different. The bottom right plot in Fig. 8, for example, shows how the left sides of the peaks are shallower as the gust approaches, and the right sides of the peaks are steeper indicating the faster change in induced AoA as the gust crosses the plate. Fig. 7 shows the k=0.3 A=30° gust canvas with the induced AoA overlaid on top of it. This figure illustrates how in the space between vortices where the gust is approaching the plate, there is a shallower change in induced AoA, and how within the bounds of the gust there is a much sharper change. Fig. 8 and Fig. 9 show how the 0.1 and 0.2 cases still contain peaks from where the vortex crosses the plate, but these are a bit harder to distinguish from the several other smaller vortices that cause higher frequency oscillations in the induced angle of attack over time. The close sets of alternating vortices present in the k=0.2 cases are also manifested in these induced AoA histories. While the k=0.3 cases have only a single peak, the 0.2 cases have a large initial peak followed by some smaller oscillations in accordance with the two following counter-rotating vortices seen in Fig. 5.

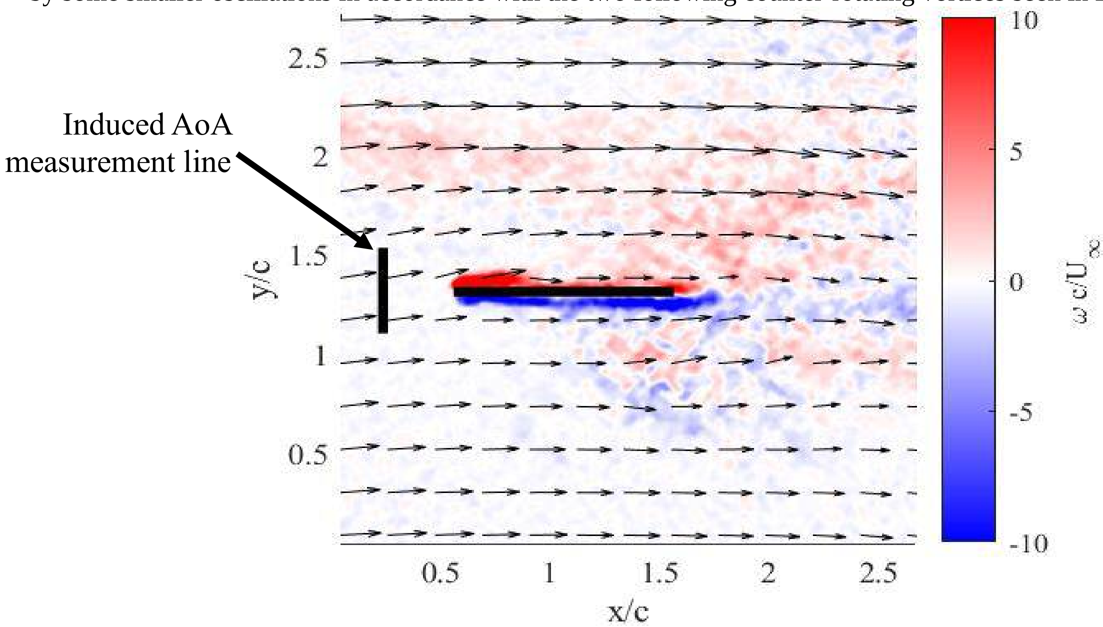

Fig. 6 Measurement line for induced AoA

$$\alpha_{induced} = \tan^{-1}\left(\frac{v}{u}\right) \quad (3)$$

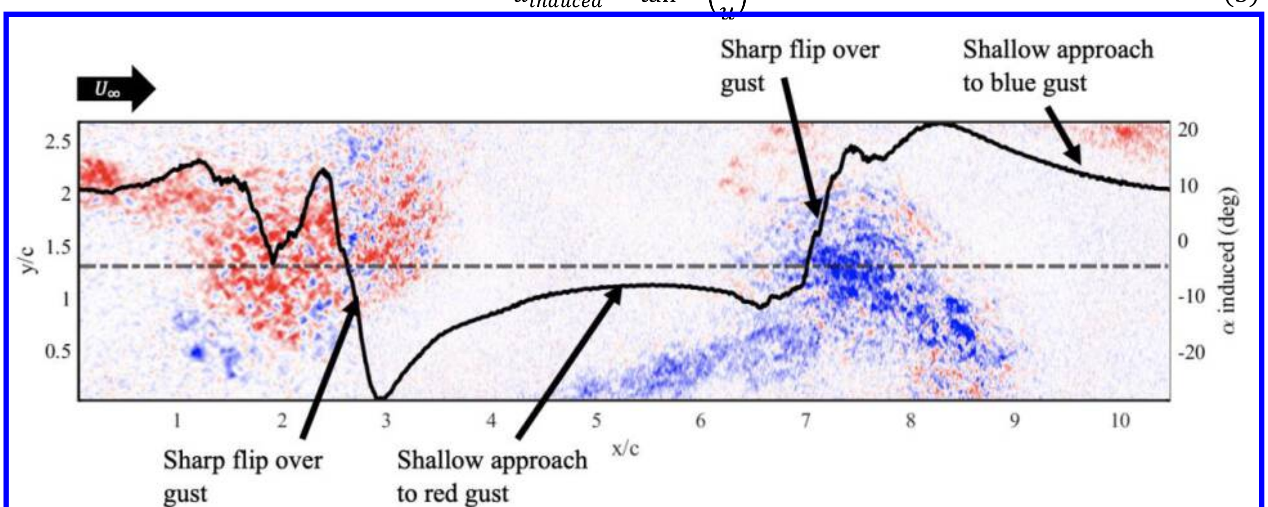

Fig. 7 k=0.3 A=30 gust canvas and induced AoA

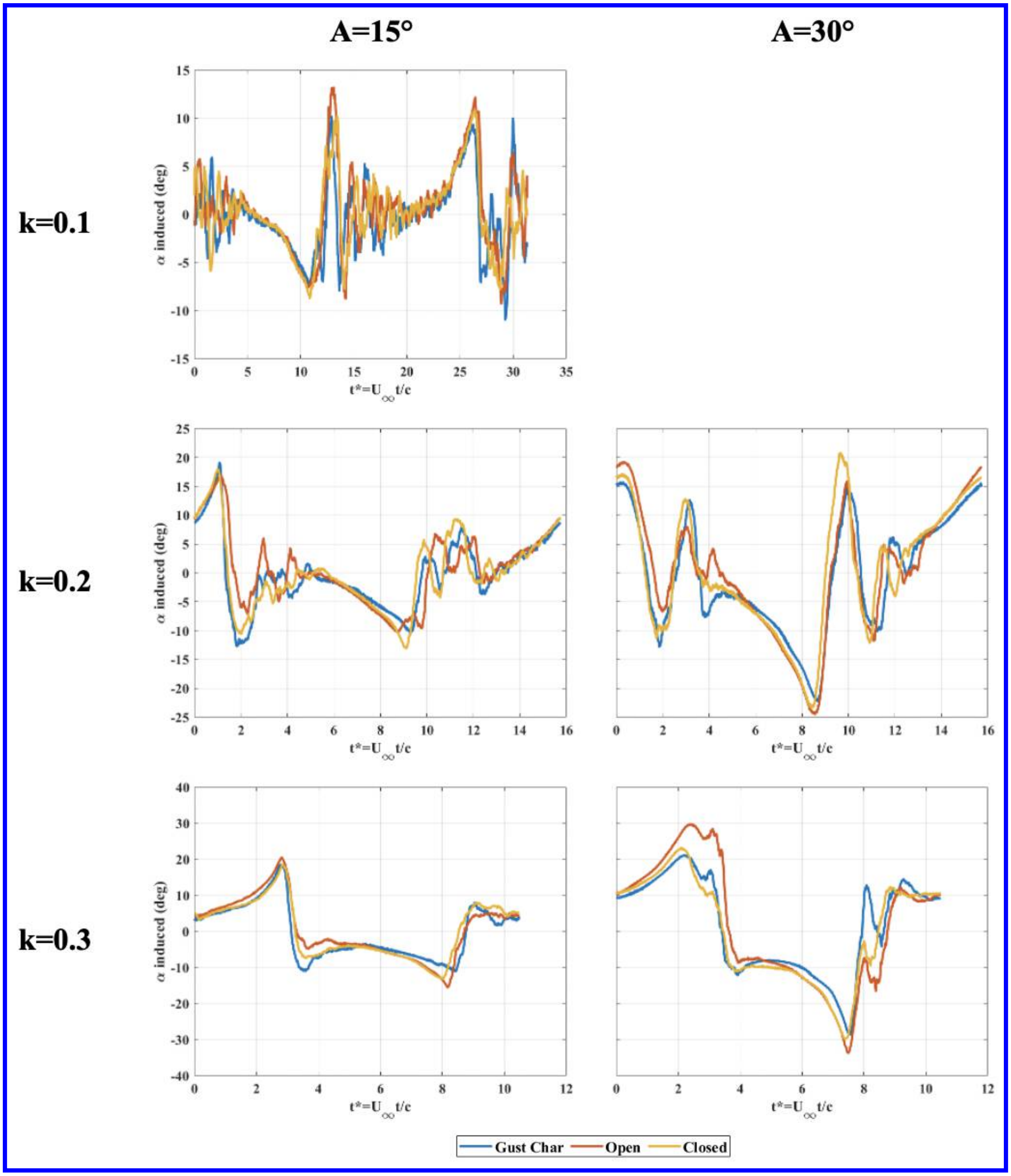

Fig. 8 Induced AoA for $C_{L_{Set=0}}$ cases (α=0° & CL=0)

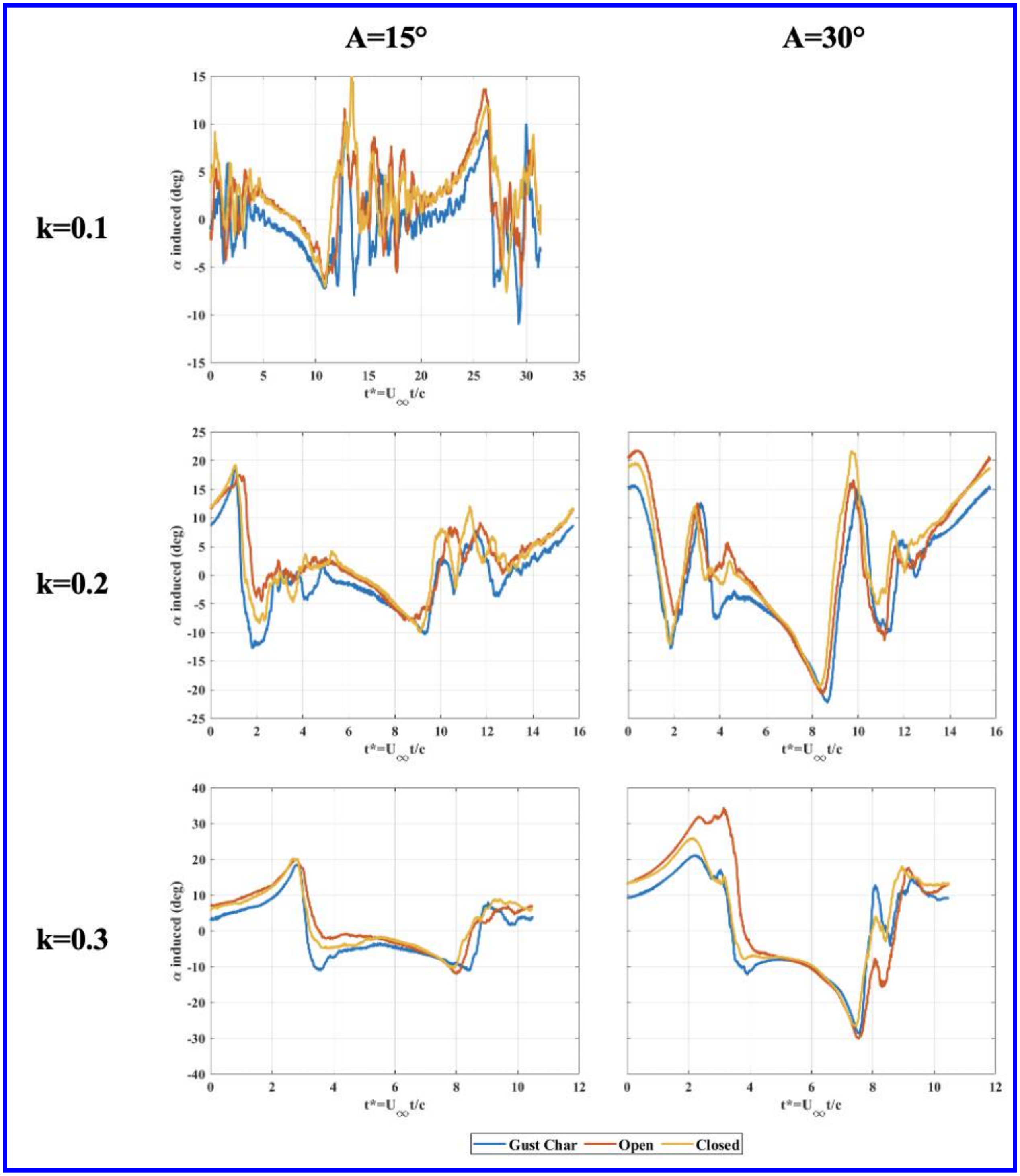

Fig. 9 Induced AoA for $C_{L_{Set\neq0}}$ cases (α=6° & C_L=0.35)

These induced angle of attack plots also show a temporally synchronized and repeatable gust across the gust characterization studies and both open and closed loop experiments. Especially in the coherent peaks, there are only small differences between each type of test with a few key patterns. For all the cases considered in Fig. 8Fig. 9, the order of appearance and the trends remain the same. The gust characterization, open loop, and closed loop are represented by the blue, red, and yellow lines, respectively and appear in that order in almost all the cases. The open loop shows the highest deviation because of the influence of the plate on the vertical velocity upstream. For example, in Fig. 8 the bottom left k=0.3 A=15° case at about t*=3.5, we can see a negative peak that occurs about when the vortex core is aligned with the LE of the plate. At this peak the signals are ordered from top to bottom as gust characterization, closed, then open loop. This order of gust characterization, closed, then open loop indicates that the closed loop controller navigates the plate through the vortices in such a way that it reduces the influence of the plate on the flow around it. This will be shown in more detail as the open and closed loop cases are compared in with PIV data. For now, we can be confident from the similarities in induced AoA that the gust is repeatable across the characterization, open, and closed loop test types and the responses of the plate can be accurately compared.

B. Mitigation

The extent of gust mitigation through closed loop control was quantified by two methods: peak to peak error and average error between the open loop and closed loop gust profiles. Peak to peak mitigation was measured by taking the highest and lowest C_L values away the target in the phase averaged period and calculating the percent reduction from open to closed loop with Eq. (4). This reduction was recorded for both positive and negative peaks then the mean of the two was taken in Table 2. Average mitigation was measured by calculating the average error through time between the target C_L value and the actual sensed C_L using Eq. (5). Again, the percent reduction from the open loop error to closed loop error was calculated in Eq. (6) to find the average mitigation. A summary of the mitigations achieved at each test point is shown in Table 2.

$$mit_{peak2peak} = 1 - \frac{(C_{L_{peak\,closed}} - C_{L_{target}})}{(C_{L_{peak\,open}} - C_{L_{target}})} \tag{4}$$

$$E_{avg} = \frac{1}{n}\sum \left| C_{L_{target}} - C_{L_{sensed}} \right| \tag{5}$$

$$mit_{avg} = 1 - \frac{E_{avg\;closed}}{E_{avg\;open}} \tag{6}$$

Table 2 Mitigation Summary

RF	Amplitude	CL Target	Peak to Peak mitigation (%)			Average Mitigation (%)
			Positive Side	Negative Side	Mean	
0.3	30	0.00	36.54	65.63	51.09	59.89
		0.35	48.29	52.79	50.54	61.02
	15	0.00	58.36	44.56	51.46	60.09
		0.35	63.59	36.93	50.26	63.32
0.2	30	0.00	36.08	48.27	42.18	50.62
		0.35	30.31	31.7	31.01	53.77
	15	0.00	54.08	21.39	37.74	50.53
		0.35	59.84	11.24	35.54	54.35
0.1	15	0.00	47.35	35.68	41.52	48.43
		0.35	60.08	31.88	45.98	55.13
Composite Average			49.45	38.01	43.73	55.72

The table shows that average mitigation is usually about 10% higher than peak to peak. That said, the mean peak to peak error still shows a significant reduction in lift with a 43.73% composite average across all test points. When considering the whole period, the composite average mitigation is even better at 55.72%. The best mitigation occurs at k=0.3 at about 50% for peak to peak and 60% for average. Fig. 10 shows how there is a dip in both mitigation types and across amplitudes at k=0.2. This is due to the aforementioned three closely grouped vortices visible in Fig. 5 that cause the oscillations in induced AoA shown in Fig. 8 and Fig. 9.

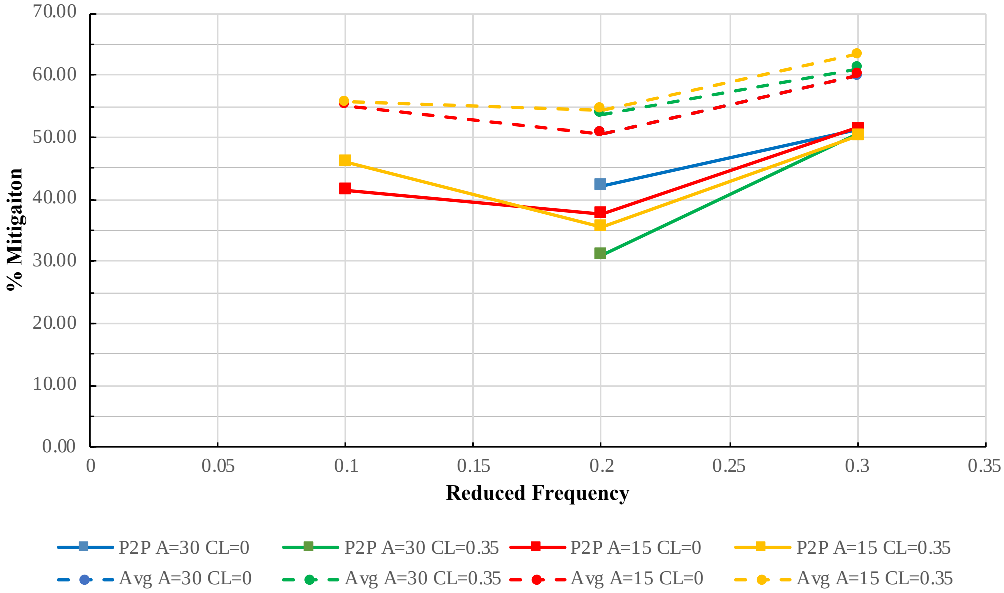

Fig. 10 Mitigation vs Reduced Frequency

A collection of individual PIV frames along with the C_L histories in open and closed loop is shown in Fig. 11. PIV frames for the open loop case where the plate was held at a constant 0° AoA are shown in the top row, and frames for the closed loop case where a C_L of zero was commanded are in the bottom row. The gray vertical lines in the central plot represent the corresponding convective times in the C_L history where these images were captured. The first frames at t*=2.72 show where the controller performs best. Here there is a large positive peak in the open loop C_L as the blue negative gust approaches the plate. In open loop, this results in a large increase in positive vorticity on the top surface of the plate indicating flow separation. Recall that this point corresponds to the peak induced angle of attack at the right edge of the blue vortex shown in Fig. 7. The closed loop case, on the other hand, has very effectively coped with the disturbance at this point, holding a C_L of 0.17 opposed to the open loop C_L 0.97, an 82% reduction.

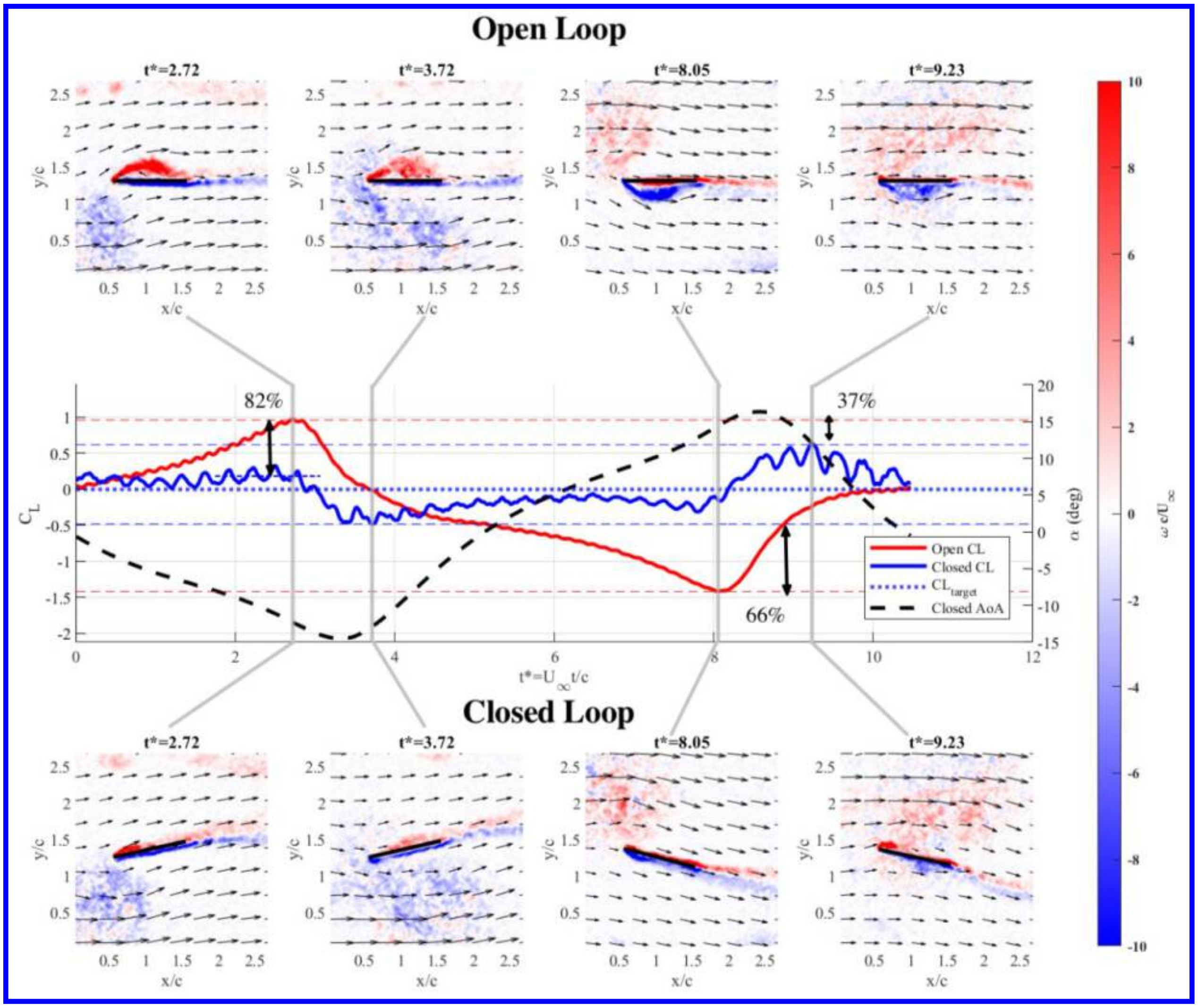

Fig. 11 k=0.3 A=30° $C_{L_{Set=0}}$ (α=0° & C_L=0) Mitigation

After the positive open loop peak, the gust begins to move across the flat plate, rapidly changing the effective AoA. By the next frame at t*=3.72, Fig. 12 shows how the orange induced AoA has reduced quickly and significantly, but the geometric AoA determined by the controller still shows a negative AoA as indicated by the dashed black line. This large negative geometric AoA is no longer appropriate at this lower induced AoA. This causes the effective AoA in yellow to be even lower and consequently the plate experiences the negative peak in the closed loop C_L at this point. Note that the C_L has been filtered with a 5Hz low pass Butterworth filter in this figure for easier viewing. The controller has sensed the negative C_L and started increasing its geometric AoA by this time, but it was too late to prevent the secondary C_L peak. This connection between quick changes in effective AoA and C_L error is further illustrated in Fig. 13 where the time derivative of effective AoA, $\dot{\alpha}_{\mathrm{eff}}$ shown by the yellow dotted line, is plotted with the solid blue C_L error. Here we can see that whenever there is a peak in $\dot{\alpha}_{\mathrm{eff}}$ there is a subsequent peak in C_L error caused by the quick change. The same trends in mitigation are shown on the opposite side of the plate in the second two frames of Fig. 11. This time, effective mitigation of a large negative peak at t*=8.05, but still a smaller subsequent positive peak at t*=9.23.

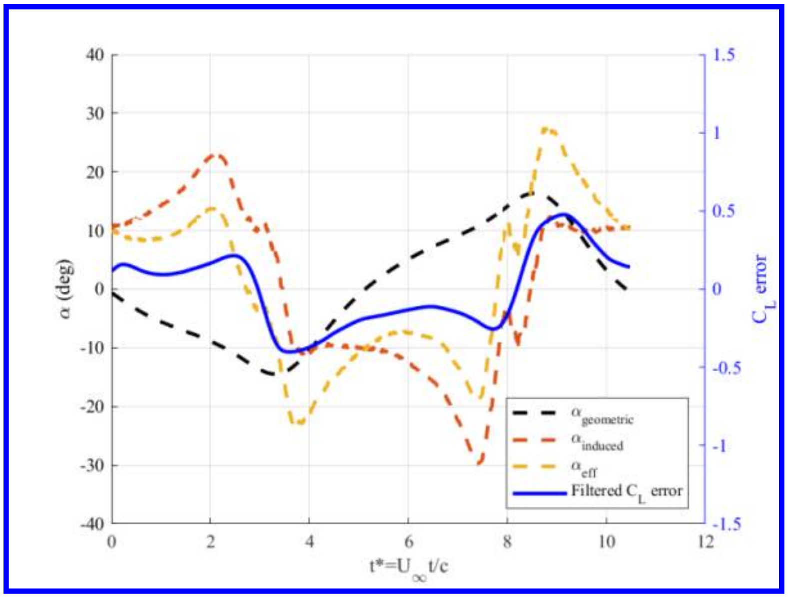

Fig. 12 Closed Loop k=0.3 A=30° C_L=0 AoAs and C_L error

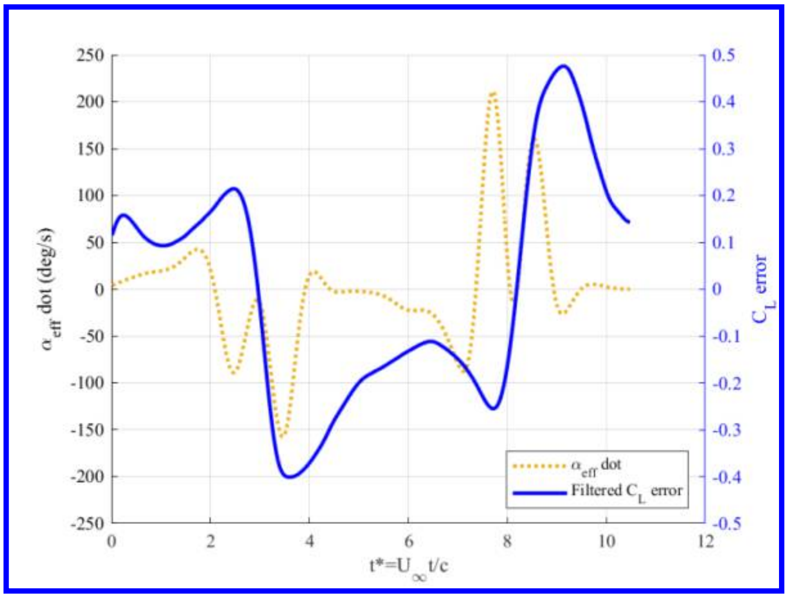

Fig. 13 Closed Loop k=0.3 A=30° CL=0 effective AoAdot and C_L error

Also visible in the 2nd and 4th closed loop frames at t*=3.72 and 9.23 is the formation of a small LEV. The frames have been enlarged and the vortices annotated in Fig. 14. This LEV is likely caused by the movement of the plate as it is not present in open loop. The vortex itself seems to only have a small effect on the forces the plate experiences but it does show that moving the plate at much higher rates could introduce further unsteady complexity to the flow such as leading and trailing edge vortices. Attempts to increase mitigation by increasing the pitch rate will have to consider the formation of these structures which may add additional influence on the C_L in addition to additional phase lag introduced by faster pitch rates. Future efforts may instead benefit from additional control methods that can compensate for global flow effects even after the gust has passed over the wing to hasten response and relaxation times.

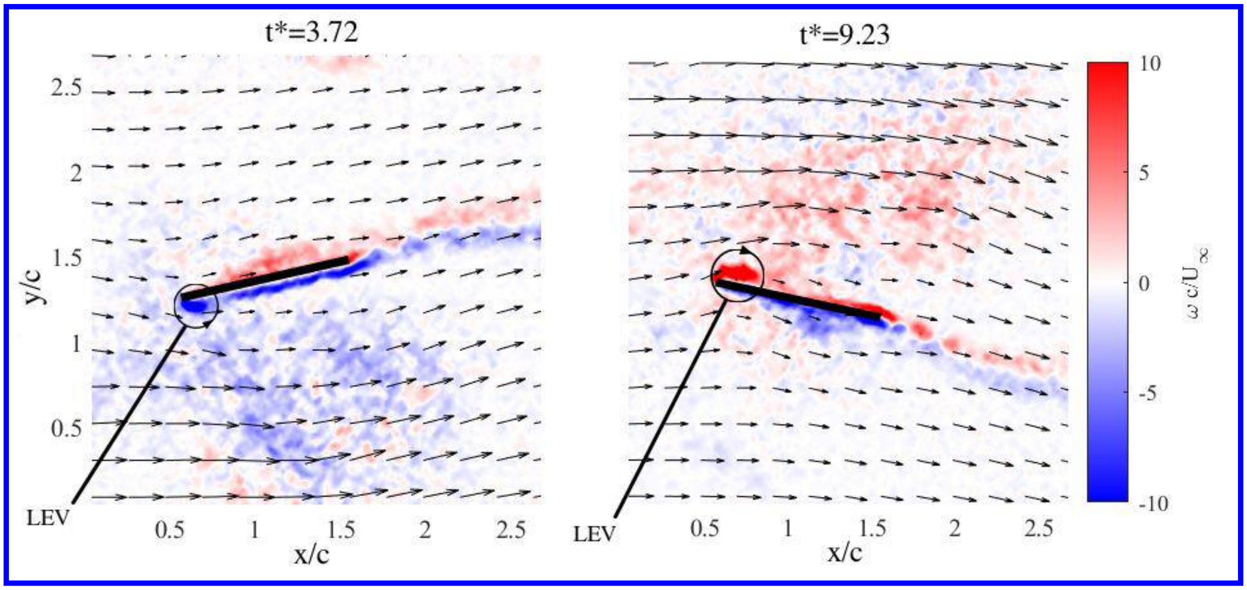

Fig. 14 LEVs in closed loop PIV frames

Similar trends are present for the k=0.2 A=30° $C_{L_{Set=0}}$ case shown in Fig. 15 with minor exceptions due to the gust coherency. In the first frames at the open loop C_L peak, we again see even more effective mitigation at 87% for the initial gust encounter as expected for the lower reduced frequency. The secondary lift peak, however, is more severe due to the group of three vortices we saw in Fig. 5. The leading negative vortex in the pair caused the first drop in induced AoA at about t*=2 shown in Fig. 16. This compounds with the geometric AoA as we saw in the k=0.3 case above. Just like the k=0.3 case, a subsequent negative peak in the closed loop C_L occurs at t*=2.19. Fig. 16 shows how the induced AoA sign flip then occurs again as the second positive vortex in the pair causes a quick increase in induced AoA at about t*=3. This generates a smaller positive C_L peak, but its magnitude is not as drastic because the geometric angle of attack is close to 0° by this point. Finally, the induced AoA changes one final time to end at about 0° as the negative third vortex crosses the plate at about t*=4. After this sequence of quick disturbances, the controller then returns to a well-controlled mitigation of the next positive-leading gust show in the next two frames.

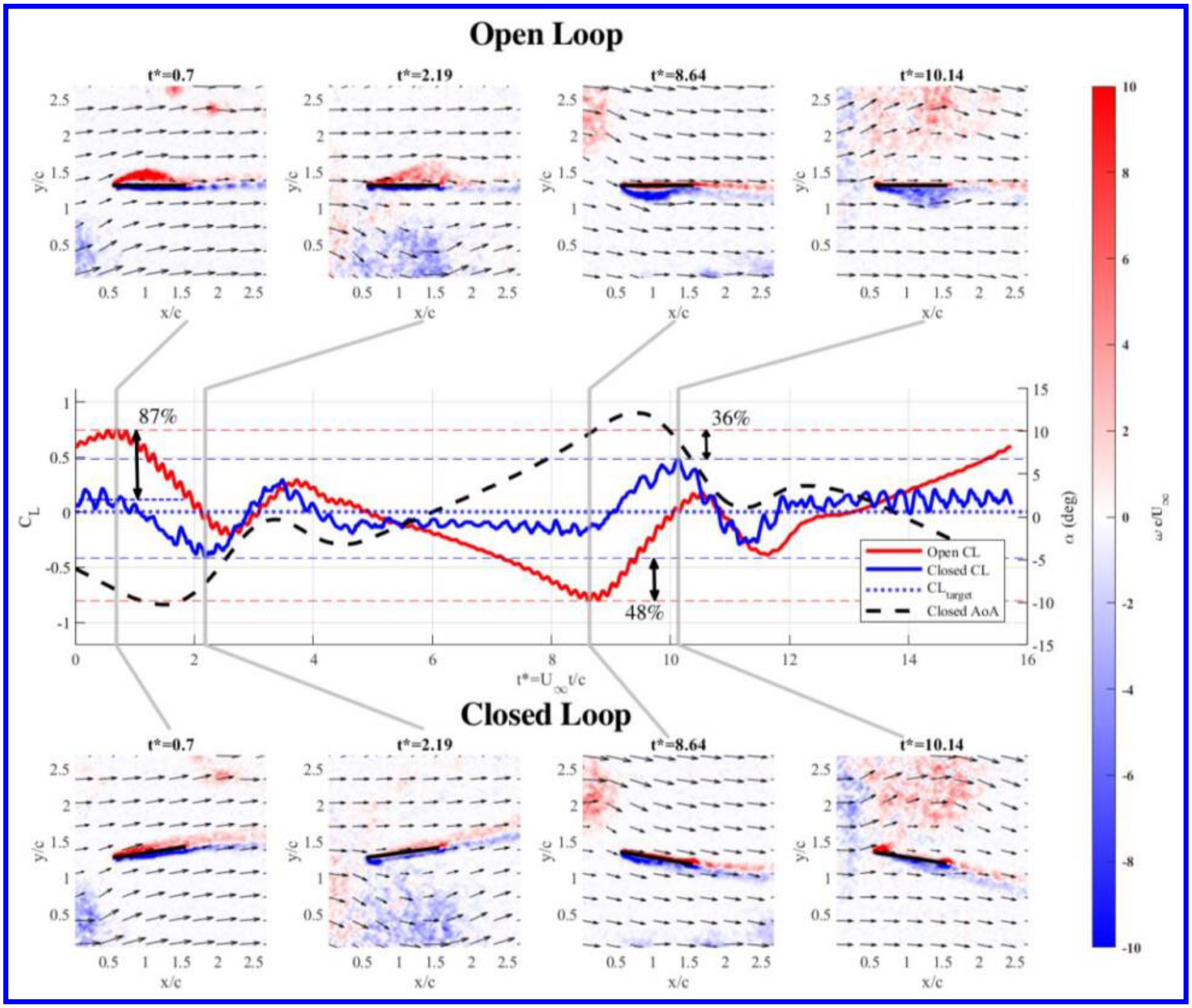

Fig. 15 k=0.2 A=30° $C_{L_{Set=0}}$ (α=0° & C_L=0 Mitigation

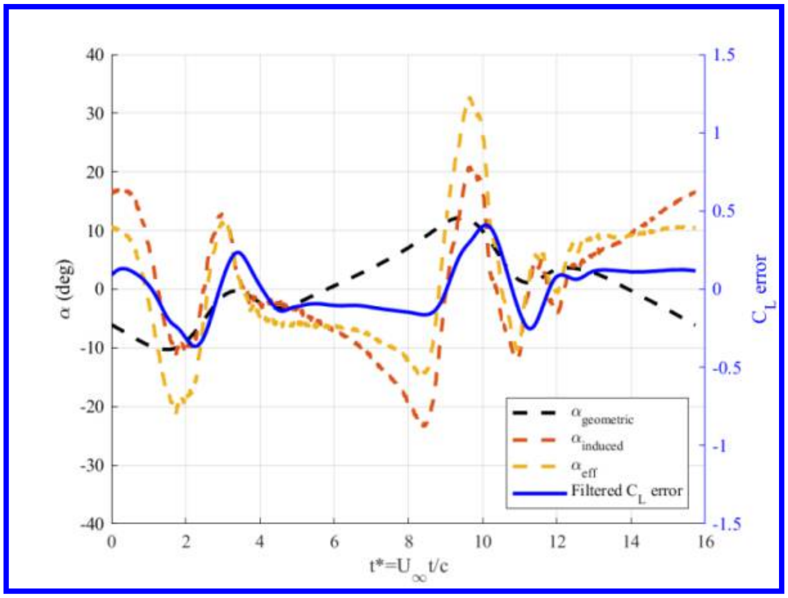

Fig. 16 Closed Loop k=0.2 A=30 C_L=0 AoAs and C_L

The connection between a large $\dot{\alpha}_{eff}$ and closed loop C_L error is again shown by the peaks in Fig. 17. Here we can see how peaks in $\dot{\alpha}_{eff}$ are followed by peaks in C_L error because the controller could not adjust quicky enough to the rapid change in the vertical velocity direction. That said, these short transients only represent a small portion of the whole period. As Fig. 15 shows, the closed loop C_L stays very close to the target C_L of 0 for much of the cycle while in open loop it deviates significantly. This results in the relatively lower peak to peak mitigation performance of 42.18% for this case, but comparable average mitigation of 50.62% shown in Table 2.

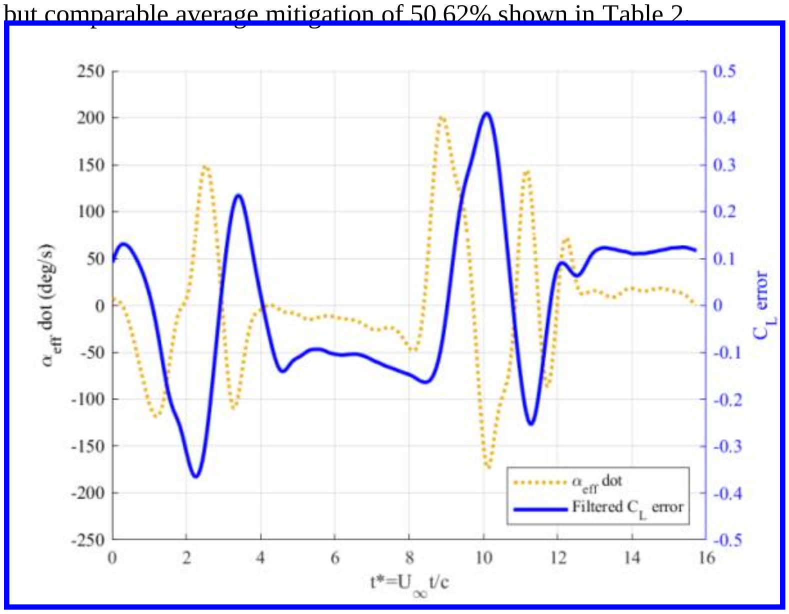

Fig. 17 Closed Loop k=0.2 A=30 C_L=0 effective AoAdot and C_L

The final case that will be discussed in detail for this paper is at k=0.3, A=15°, and C_L=0.35 shown in Fig. 18. This case differs from the previous with the 15° amplitude and $C_{L_{Set\neq 0}}$ at a C_L=0.35 and α=6 ° rather than 0. As discussed in the Gust Characterization Section, the 15° amplitude results in both a smaller gust as shown in Fig. 5, and less deviation from the centerline. In Fig. 11 and Fig. 15 where A=30°, the PIV 1st and 3rd frames show how the vortices approach the flat plate slightly above or below, but Fig. 18 shows how at A=15° the vortices are more centered on the plate as they approach.

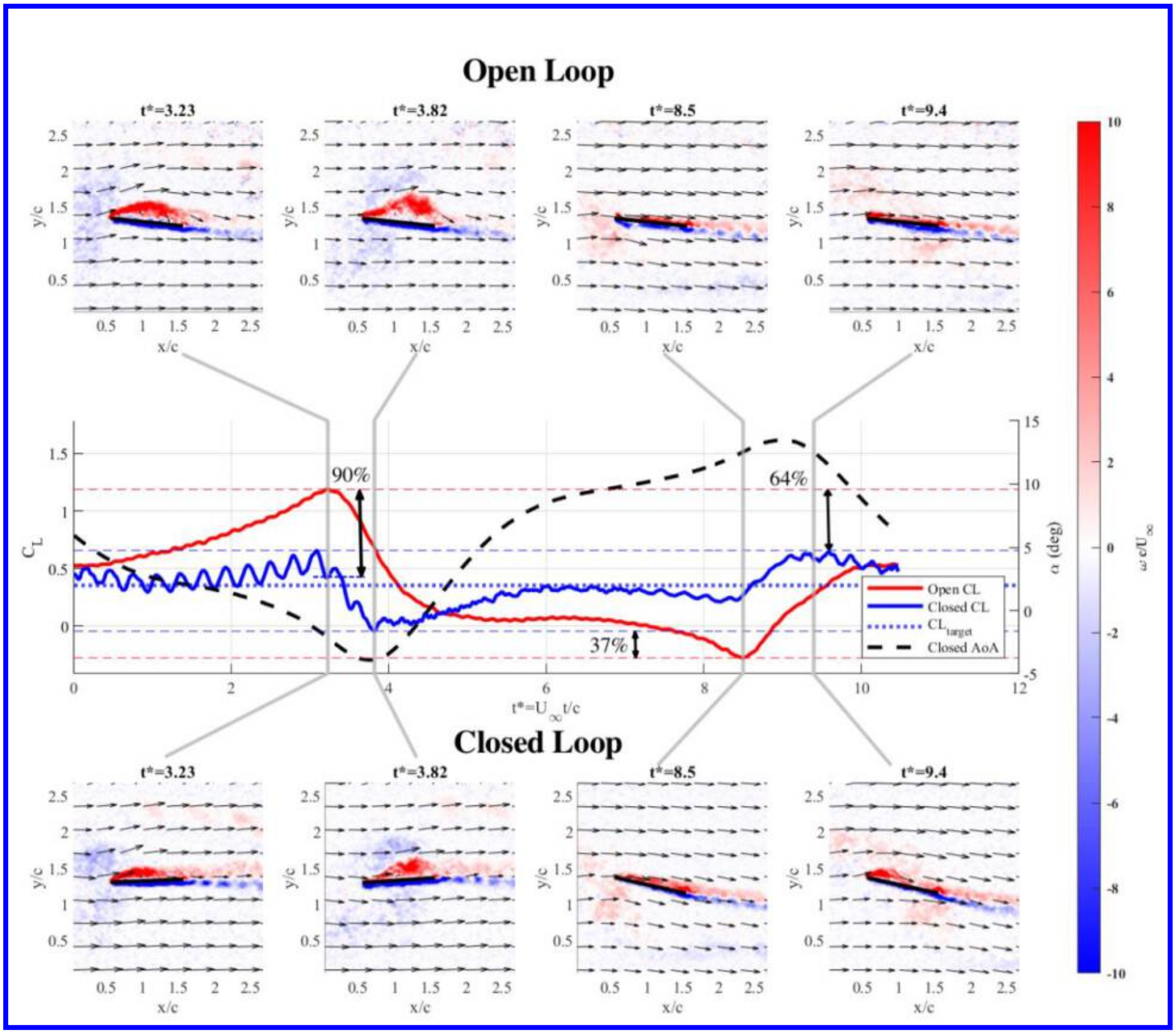

Fig. 18 k=0.3, A=15, $C_{L_{Set \neq 0}}$ (α=6° & C_L=0.35) Mitigation

Commanding a nonzero C_L also resulted in several key differences from the other two cases. Commanding a positive α and C_L shifted both the open and closed loop C_L histories up to be more centered around 0.35. Holding a positive C_L resulted in a visually smaller reduction in upper surface vorticity from open to closed loop at t*=3.23 where the first open loop C_L peak occurs. There is still very effective 90% mitigation at this initial peak which shows that the controller can also achieve a nonzero C_L at this point in a gust encounter. There is still a subsequent negative peak, however, as the vortex crosses the LE of the plate flipping the induced angle of attack from positive to negative as shown by Fig. 19.

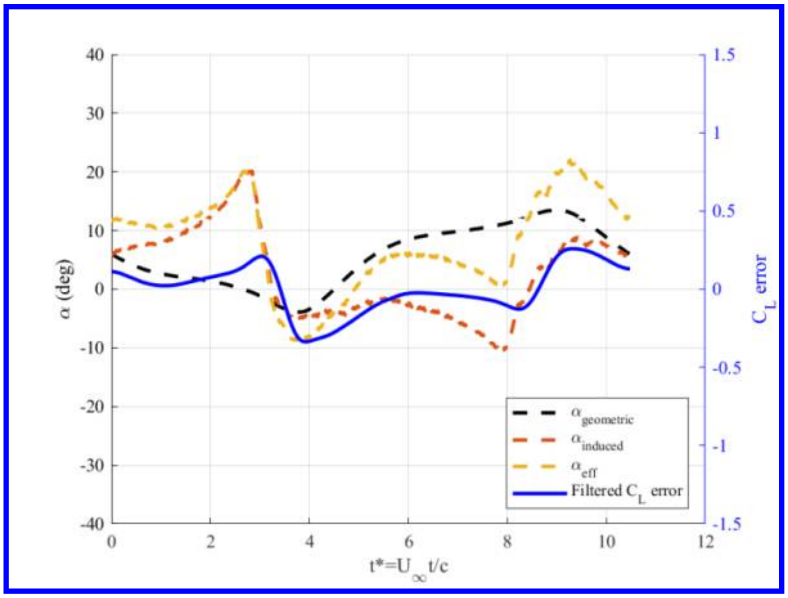

Fig. 19 k=0.3 A=15 CL=0.3 Closed Loop AoAs and C_L error

IV.Conclusion

Vortical gust mitigation has been performed using a closed loop controller while time synchronized force and TR-PIV measurements were collected to study the flow physics present during successful mitigation. A flat plate with an attached force transducer sensed the lift force which was fed to a PID-based closed loop controller to determine the AoA needed to achieve a commanded C_L. A variety different amplitude and reduced frequency periodic vortical gusts were generated using a pitching flat plate gust generator. These gusts were successfully mitigated by the closed loop controller up to 63.32% when comparing open loop C_L deviations to closed loop. The controller achieved effective mitigation of around 50% through a variety of gust profiles with varying coherency and location relative to the plate depending on the reduced frequency and gust generator amplitude. These experiments show that effective mitigation can be even achieved through closed loop control without knowledge of the aerodynamics of the plant.

PIV measurements revealed that the controller achieved the most effective mitigation while the gust approached the flat plate. During this phase, relatively gradual changes in induced AoA imparted by the gust could be mitigated by pitching the flat plate accordingly. PIV results showed that while there was a large increase in vorticity on one side of the plate in open loop, the closed loop controller reduced this vorticity better maintaining flow attachment and reducing C_L deviations through active AoA actuation.

The controller struggled when the vortex gust would cross the LE of the plate, rapidly flipping the sign of the induced AoA. The controller could not react fast enough to completely negate this flow change which resulted in a subsequent smaller peak in the closed loop C_L as the controller caught up. This smaller subsequent peak was also accompanied by the formation of a LEV as the controller changed direction of the geometric AoA to match the new induced AoA. The rapid change in induced AoA as the vortex crossed the plate is inherent to the structure of a vortical gust and is what makes their mitigation uniquely challenging. A focus on the rapid sensing and actuation required during this phase will be important to achieve effective mitigation throughout an entire periodic vortical gust encounter.

These experiments indicate that effective mitigation is achieved when flow separation is limited, in this case by pitching to minimize effective AoA. To achieve higher mitigations, the controller must be able to sense and respond incoming disturbances faster. A feedforward control system that can sense either the upstream velocity or induced AoA could be very effective in informing the controller ahead of time what an appropriate response would be to the incoming disturbance without necessarily knowing its whole structure a priori. While pitching has shown admirable mitigation capabilities, the formation of LEVs during some of the faster pitching maneuvers may indicate that increasing the pitch rate further could introduce greater complexity to the flow which may add additional challenges to control efforts. An additional actuation method such as wing morphing to change camber is an interesting prospect to rapidly change C_L while limiting influence on the flow around the plate. Active flow control techniques such as plasma actuators implemented on a vertical axis wind turbine by Ronen and Greenblatt [18] also present an attractive way of controlling flow separation caused by vortical gust encounters.

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